

H_0 and conclude that there is no significant difference.

5.5. LEVEL OF SIGNIFICANCE AND POWER OF TEST

Probability of type I error is called level of significance or significance level.

i.e., Significance level = Probability [Rejecting H_0/H_0 is true]

Probability of rejecting H_0 when the alternative hypothesis H_1 is true is called the power of the test.

$$\begin{aligned}\text{Power} &= \Pr [\text{Rejecting } H_0/H_1 \text{ is true}] \\ &= 1 - \Pr [\text{Accepting } H_0/H_1 \text{ is true}] \\ &= 1 - \Pr [\text{Type II error}].\end{aligned}$$

The best test is that for which the significance level is less than a preassigned small number (α) and power is maximum.

ILLUSTRATIVE EXAMPLES

Example 1. A population follows normal distribution with mean μ and variance 9. To test $H_0: \mu = 5$ against $\mu = 7$, the test procedure suggested is to reject H_0 if $\bar{x} \geq 6$ where $\bar{x}$ is the mean of a sample of size 16. Find the significance level and the power of the test.

(CUSAT 2012)

Sol. $H_0 : \mu = 5$ V_s $H_1 : \mu = 7, \sigma = 3, n = 16$

Significance level = P [Rejecting H_0/H_0 is true]

$$= P [\bar{x} \geq 6/\mu = 5]$$

$$= P \left[\frac{\bar{x} - \mu}{\sigma/\sqrt{n}} \geq \frac{6 - \mu}{\sigma/\sqrt{n}} \middle| \mu = 5 \right]$$

$$= P \left[Z \geq \frac{6 - 5}{3/\sqrt{16}} \right] = P \left[Z \geq \frac{4}{3} \right]$$

$$= 0.5 - 0.4082 = \mathbf{0.0918}$$

Power of test = P [Rejecting H_0/H_1 is true]

$$= P [\bar{x} \geq 6/\mu = 7]$$

$$= P \left[\frac{\bar{x} - \mu}{\sigma/\sqrt{n}} \geq \frac{6 - \mu}{\sigma/\sqrt{n}} \middle| \mu = 7 \right] = P \left[Z \geq \frac{6 - 7}{3/\sqrt{16}} \right]$$

$$= P [Z \geq -1.33] = 0.5 + 0.4082 = \mathbf{0.9082}.$$

Example 2. A continuous random variable X has a frequency function $f(x) = \frac{1}{\theta}, 0 < x < \theta$. It is desired to test $H_0 : \theta = 1$ V_s $H_1 : \theta = 2$ using a single observation X and $X \geq 0.95$ is used as critical region. Evaluate of type I and II errors. (CUSAT 2010)

Sol. $H_0 : \theta = 1$ V_s $H_1 : \theta = 2$ given that C.R. is $X \geq 0.95, f(x) = \frac{1}{\theta}, 0 < x < \theta$

Pro [Type I error] = P [Rejecting H_0/H_0 is true]

$$= P [X \geq 0.95/\theta = 1]$$

$$= \int_{0.95}^1 f(x) dx / \theta = 1$$

$$= \int_{0.95}^1 1 dx = 0.05$$

P [Type II error] = P [Accepting H_0/H_0 is false]

$$= P [x \leq 0.95/\theta = 2] = \int_0^{0.95} \frac{1}{2} dx = \mathbf{0.475}$$

21.82. CHI-SQUARE (χ^2) TEST

When a coin is tossed 200 times, the theoretical considerations lead us to expect 100 heads and 100 tails. But in practice, these results are rarely achieved. The quantity χ^2 (a Greek letter, pronounced as chi-square) describes the magnitude of discrepancy between theory and observation. If $\chi = 0$, the observed and expected frequencies completely coincide. The greater the discrepancy between the observed and expected frequencies, the greater is the value of χ^2 . Thus χ^2 affords a measure of the correspondence between theory and observation.

If O_i ($i = 1, 2, \dots, n$) is a set of observed (experimental) frequencies and E_i ($i = 1, 2, \dots, n$) is the corresponding set of expected (theoretical or hypothetical) frequencies, then, χ^2 is defined as

$$\chi^2 = \sum_{i=1}^n \left[\frac{(O_i - E_i)^2}{E_i} \right]$$

where $\sum O_i = \sum E_i = N$ (total frequency) and degrees of freedom ($d.f.$) = $(n - 1)$.

Note. (i) If $\chi^2 = 0$, the observed and theoretical frequencies agree exactly.

(ii) If $\chi^2 > 0$ they do not agree exactly.

21.82.1. Degree of Freedom

While comparing the calculated value of χ^2 with the table value, we have to determine the degrees of freedom.

If we have to choose any four numbers whose sum is 50, we can exercise our independent choice for any three numbers only, the fourth being 50 minus the total of the three numbers selected. Thus, though we were to choose any four numbers, our choice was reduced to three because of one condition imposed. There was only one restraint on our freedom and our degrees of freedom were $4 - 1 = 3$. If two restrictions are imposed, our freedom to choose will be further curtailed and degrees of freedom will be $4 - 2 = 2$.

In general, the number of degrees of freedom is the total number of observations less the number of independent constraints imposed on the observations. Degrees of freedom ($d.f.$) are usually denoted by ν (the letter 'nu' of the Greek alphabet).

Thus, $\nu = n - k$, where k is the number of independent constraints in a set of data of n observations.

Note. (i) For a $p \times q$ contingency table (p columns and q rows), $\nu = (p - 1)(q - 1)$

(ii) In the case of a contingency table, the expected frequency of any class

$$= \frac{\text{Total of row in which it occurs} \times \text{Total of columns in which it occurs}}{\text{Total number of observations}}$$

degrees of freedom are less than the total frequency, mean and standard deviation.

[If the data is given in a series of " n " number then degrees of freedom = $n - 1$.

[In the case of Binomial distribution d.f. = $n - 1$

In the case of Poisson distribution d.f. = $n - 2$

In the case of Normal distribution d.f. = $n - 3$.]

21.82.4. χ^2 Test as a Test of Goodness of Fit

χ^2 test enables us to ascertain how well the theoretical distributions such as Binomial, Poisson or Normal etc. fit empirical distributions, *i.e.*, distributions obtained from sample data.

If the calculated value of χ^2 is less than the table value at a specified level (generally 5%) of significance, the fit is considered to be good i.e., the divergence between actual and expected frequencies is attributed to fluctuations of simple sampling. If the calculated value of χ^2 is greater than the table value, the fit is considered to be poor.

ILLUSTRATIVE EXAMPLES

Example 1. The following table gives the number of accidents that took place in an industry during various days of the week. Test if accidents are uniformly distributed over the week.

Day	Mon	Tue	Wed	Thu	Fri	Sat
No. of accidents	14	18	12	11	15	14

Sol. Null hypothesis H_0 . The accidents are uniformly distributed over the week.

Under this H_0 , the expected frequencies of the accidents on each of these days = $\frac{84}{6} = 14$

Observed frequency O_i	14	18	12	11	15	14
Expected frequency E_i	14	14	14	14	14	14
$(O_i - E_i)^2$	0	16	4	9	1	0

$$\chi^2 = \frac{\sum(O_i - E_i)^2}{E_i} = \frac{30}{14} = 2.1428.$$

Conclusion. Table value of χ^2 at 5% level for $(6 - 1 = 5 \text{ d.f.})$ is 11.09.

Since the calculated value of χ^2 is less than the tabulated value, H_0 is accepted i.e., the accidents are uniformly distributed over the week.

Example 2. A die is thrown 276 times and the results of these throws are given below :

No. appeared on the die	1	2	3	4	5	6
Frequency	40	32	29	59	57	59

Test whether the die is biased or not.

Sol. Null hypothesis H_0 . Die is unbiased.

Under this H_0 , the expected frequencies for each digit is $\frac{276}{6} = 46$

To find the value of χ^2

O_i	40	32	29	59	57	59
E_i	46	46	46	46	46	46
$(O_i - E_i)^2$	36	196	289	169	121	169

$$\chi^2 = \frac{\Sigma(O_i - E_i)^2}{E_i} = \frac{980}{46} = 21.30.$$

Conclusion. Tabulated value of χ^2 at 5% level of significance for $(6 - 1 = 5)$ d.f. is 11.09. Since the calculated value of $\chi^2 = 21.30 > 11.07$ the tabulated value, H_0 is rejected. i.e., die is not unbiased or die is biased.

Example 3. The following table shows the distribution of digits in numbers chosen at random from a telephone directory :

Digits	0	1	2	3	4	5	6	7	8	9
Frequency	1026	1107	997	966	1075	933	1107	972	964	853

Test whether the digits may be taken to occur equally frequently in the directory.

Sol. Null hypothesis H_0 . The digits taken in the directory occur equally frequency. i.e., there is no significant difference between the observed and expected frequency.

Under H_0 , the expected frequency is given by $= \frac{10,000}{10} = 1000$

To find the value of χ^2

O_i	1026	1107	997	996	1075	1107	933	972	964	853
E_i	1000	1000	1000	1000	1000	1000	1107	1000	1000	1000
$(O_i - E_i)^2$	676	11449	9	1156	5625	11449	4489	784	1296	21609

$$\chi^2 = \frac{\Sigma(O_i - E_i)^2}{E_i} = \frac{58542}{1000} = 58.542.$$

Conclusion. The tabulated value of χ^2 at 5% level of significance for 9 d.f. is 16.919. Since the calculated value of χ^2 is greater than the tabulated value, H_0 is rejected.

i.e., there is significant difference between the observed and theoretical frequency.

i.e., the digits taken in the directory do not occur equally frequently.

Example 4. Records taken of the number of male and female births in 800 families having four children are as follows :

No. of male births	0	1	2	3	4
No. of female births	4	3	2	1	0
No. of families	32	178	290	236	94

Test whether the data are consistent with the hypothesis that the binomial law holds and the chance of male birth is equal to that of female birth, namely $p = q = 1/2$.

Sol. H_0 : The data are consistent with the hypothesis of equal probability for male and female births. i.e., $p = q = 1/2$.

We use binomial distribution to calculate theoretical frequency given by :

$$N(r) = N \times P(X = r)$$

where N is the total frequency. $N(r)$ is the number of families with r male children :

$$P(X = r) = {}^nC_r p^r q^{n-r}$$

where p and q are probability of male and female birth, n is the number of children.

$$N(0) = \text{No. of families with 0 male children} = 800 \times {}^4C_0 \left(\frac{1}{2}\right)^4 = 800 \times 1 \times \frac{1}{2^4} = 50$$

$$N(1) = 800 \times {}^4C_1 \left(\frac{1}{2}\right)^1 \left(\frac{1}{2}\right)^3 = 200 ; N(2) = 800 \times {}^4C_2 \left(\frac{1}{2}\right)^2 \left(\frac{1}{2}\right)^2 = 300$$

$$N(3) = 800 \times {}^4C_3 \left(\frac{1}{2}\right)^1 \left(\frac{1}{2}\right)^3 = 200 ; N(4) = 800 \times {}^4C_4 \left(\frac{1}{2}\right)^0 \left(\frac{1}{2}\right)^4 = 50$$

Observed frequency O_i	32	178	290	236	94
Expected frequency E_i	50	200	300	200	50
$(O_i - E_i)^2$	324	484	100	1296	1936
$\frac{(O_i - E_i)^2}{E_i}$	6.48	2.42	0.333	6.48	38.72

$$\chi^2 = \frac{\sum (O_i - E_i)^2}{E_i} = 54.433.$$

Conclusion. Table value of χ^2 at 5% level of significance for $5 - 1 = 4$ d.f. is 9.49.

Since the calculated value of χ^2 is greater than the tabulated value, H_0 is rejected.

i.e., the data are not consistent with the hypothesis that the binomial law holds and that the chance of a male birth is not equal to that of a female birth.

Note. Since the fitting is binomial, the degrees of freedom $\nu = n - 1$ i.e., $\nu = 5 - 1 = 4$.

Example 5. Verify whether Poisson distribution can be assumed from the data given below :

No. of defects	0	1	2	3	4	5
Frequency	6	13	13	8	4	3

Sol. H_0 : Poisson fit is a good fit to the data.

$$\text{Mean of the given distribution} = \frac{\sum f_i x_i}{\sum f_i} = \frac{94}{47} = 2$$

To fit a Poisson distribution we require m . Parameter $m = \bar{x} = 2$.

By Poisson distribution the frequency of r success is

$$N(r) = N \times e^{-m} \cdot \frac{m^r}{r!}, \text{ } N \text{ is the total frequency.}$$

$$N(0) = 47 \times e^{-2} \cdot \frac{(2)^0}{0!} = 6.36 = 6; N(1) = 47 \times e^{-2} \cdot \frac{(2)^1}{1!} = 12.72 \approx 13$$

$$N(2) = 47 \times e^{-2} \cdot \frac{(2)^2}{2!} = 12.72 \approx 13; N(3) = 47 \times e^{-2} \cdot \frac{(2)^3}{3!} = 8.48 \approx 9$$

$$N(4) = 47 \times e^{-2} \cdot \frac{(2)^4}{4!} = 4.24 \approx 4; N(5) = 47 \times e^{-2} \cdot \frac{(2)^5}{5!} = 1.696 \approx 2.$$

X	0	1	2	3	4	5
O_i	6	13	13	8	4	3
E_i	6.36	12.72	12.72	8.48	4.24	1.696
$\frac{(O_i - E_i)^2}{E_i}$	0.2037	0.00616	0.00616	0.02716	0.0135	1.0026

$$\chi^2 = \frac{\sum (O_i - E_i)^2}{E_i} = 1.2864.$$

Conclusion. The calculated value of χ^2 is 1.2864. Tabulated value of χ^2 at 5% level of significance for $\gamma = 6 - 2 = 4$ d.f. is 9.49. Since the calculated value of χ^2 is less than that of tabulated value, H_0 is accepted i.e., Poisson distribution provides a good fit to the data.

Example 6. The theory predicts the proportion of beans in the four groups, G_1, G_2, G_3, G_4 should be in the ratio 9 : 3 : 3 : 1. In an experiment with 1600 beans the numbers in the four groups were 882, 313, 287 and 118. Does the experimental result support the theory.

Sol. H_0 . The experimental result support the theory. i.e., there is no significant difference between the observed and theoretical frequency under H_0 , the theoretical frequency can be calculated as follows :

$$E(G_1) = \frac{1600 \times 9}{16} = 900; E(G_2) = \frac{1600 \times 3}{16} = 300;$$

$$E(G_3) = \frac{1600 \times 3}{16} = 300; E(G_4) = \frac{1600 \times 1}{16} = 100$$

To calculate the value of χ^2 .

Observed frequency O_i	882	313	287	118
Expected frequency E_i	900	300	300	100
$\frac{(O_i - E_i)^2}{E_i}$	0.36	0.5633	0.5633	3.24

$$\chi^2 = \frac{\sum (O_i - E_i)^2}{E_i} = 4.7266.$$

Conclusion. Table value of χ^2 at 5% level of significance for 3 d.f. is 7.815. Since the calculated value of χ^2 is less than that of the tabulated value. Hence H_0 is accepted. i.e., the experimental result support the theory.